

Qualification Pack



Technical Helper Distribution

QP Code: PSS/Q0101

Version: 1.0

NSQF Level: 2

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PSS/Q0101: Technical Helper Distribution

Brief Job Description

The incumbent in the job will carry out equipment upkeep, material and equipment transportation, cable laying and other activities to support erection and maintenance of power poles, structures, overhead and underground powerlines and cables, etc. They will help in fixing nuts and bolts and other mechanical support to the Lineman as well.

Personal Attributes

Physically and mentally able to safely perform essential functions of the job. The candidate should be able to climb ladders, scaffolds, poles and towers of various heights. Also able to crawl and work in confined spaces such as attics, manholes and crawlspaces.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [PSS/N0101: Assisting in repair and maintenance of power distribution lines and components](#)
2. [PSS/N0102: Preparation and assistance for erection and maintenance of power distribution lines](#)
3. [PSS/N0103: Assisting in operation and maintenance of 11/0.433 KV Distribution Substation](#)
4. [PSS/N0104: Assisting in laying of underground and AB cables](#)
5. [PSS/N1336: Working effectively with others](#)
6. [PSS/N2001: Use basic health and safety practices for power related work](#)

Qualification Pack (QP) Parameters

Sector	Power
Sub-Sector	Distribution
Occupation	Distribution System Erection, Commissioning, Operation & Maintenance
Country	India
NSQF Level	2



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Credits	NA
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7245.90
Minimum Educational Qualification & Experience	8th Class
Minimum Level of Education for Training in School	Same as Min.Ed for Job
Pre-Requisite License or Training	Electrical - 6 months (Suggested but not Mandatory)
Minimum Job Entry Age	18 Years
Last Reviewed On	13/09/2017
Next Review Date	31/03/2022
NSQC Approval Date	20/07/2015
Version	1.0
Reference code on NQR	2015/POW/PSSC/00880
NQR Version	1.0



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PSS/N0101: Assisting in repair and maintenance of Power Distribution Lines and components

Description

This unit covers the competencies required by assistants who help linemen in the repair and maintenance for Power Transmission and Distribution Lines and components. This includes handling of tools and equipment for installation and maintenance and carrying out necessary tasks in a safe, efficient and effective manner.

Scope

This unit/task covers the following:

- Working safely
- Handling tools and equipment
- Assisting in repair and maintenance for Power distribution lines
- Post repair and maintenance activities

Elements and Performance Criteria

Working safely

To be competent, the user/individual on the job must be able to:

- PC1.** work safely at all times, complying with health and safety legislation, regulations and other relevant guidelines
- PC2.** adhere to procedures or systems in place for health and safety, personal protective equipment (ppe) and other relevant safety regulations for electrical and related operations
- PC3.** work following laid down procedures and instructions
- PC4.** ensure that all tools, equipment, etc. Are in a safe and usable condition and are kept at secured location
- PC5.** ensure work area is clean and safe from hazards before and after the job is completed

Handling tools and equipment for repair and maintenance

To be competent, the user/individual on the job must be able to:

- PC6.** Performance Criteria Description *
- PC7.** identify various types of circuits and its components correctly
- PC8.** identify accurately and acquire correct tools, equipment required for various aspects of repair and maintenance of distribution lines and components
- PC9.** ensure the tools and equipment are well maintained and handled, and approved for use
- PC10.** select and use tools and equipment such as chainsaws, pruning saws, manlifts and chippers to trim trees or its branches where required
- PC11.** identify hazards of trimming trees such as limits of approach, public safety and step and touch potential
- PC12.** clean and handle transmission and distribution line tools according to standard procedures

Assist in repair and maintenance of Power distribution lines



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To be competent, the user/individual on the job must be able to:

- PC13.** follow instruction, job standards and specifications accurately and safely while assisting linemen in performing off-line overhead and underground line maintenance procedure according to company procedures
- PC14.** hold, align, load and unload, poles and related equipment as per instructions and job standards and requirements accurately and safely while assisting in pole dismantling and re-setting procedure as per company procedures
- PC15.** carry out conductor stringing procedures, paving conductor on the ground along the pole under the guidance of lineman/supervisor as per instructions
- PC16.** re-tighten stay wires as per standard procedure, on instructions
- PC17.** tighten nuts and bolts securely for aligning lines and under guidance of lineman
- PC18.** hold, handover components, measure, unpack, join, align, load, unload equipment as per instructions, job standards and requirements accurately and safely while assisting linemen in repair of conductors by splicing, jointing, using armor rods, line guards and vibration dampers

Post-repair and maintenance activities

To be competent, the user/individual on the job must be able to:

- PC19.** deal promptly and effectively with problems within control, and seek help and guidance from the relevant people for problems that cannot be resolved
- PC20.** leave the work area in a safe and tidy condition on completion of the repair and maintenance activities
- PC21.** refer unresolved job related problems to appropriate personnel for support
- PC22.** monitor the problem and keep the supervisor informed about progress or any delays in resolving the problem

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** relevant legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions
- KU2.** relevant health and safety requirements applicable in the work place
- KU3.** own job role and responsibilities and sources for information pertaining to employment terms, entitlements, job role and responsibilities
- KU4.** reporting structure, inter-dependent functions, lines and procedures in the work area
- KU5.** how to engage with specialists for support in order to resolve incidents and service requests
- KU6.** importance of working in clean and safe environment practices and procedures
- KU7.** relevant people and their responsibilities within the work area
- KU8.** escalation matrix and procedures for reporting work and employment related issues
- KU9.** common electricity terminology and correct interpretation of the same Terminology: e.g. current, voltage, resistance, kilowatt (kw), kilowatt hour: (kwh)(unit of electric consumption)
- KU10.** specific terminology used in distribution line work Terminology: e.g. pole, tower, load, load shedding, etc.



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- KU11.** elements of the power system; Elements: e.g. generation, transmission, distribution, metering, etc
- KU12.** different types of material and accessories used in power T&D Materials and accessories: e.g. supports (poles-steel, cement , wooden), conductors (sizes, current carrying capacity), conductor accessories, binding tape, binding wire, p.g. clamp, t clamp etc. , insulators (pin, disc, shackle, guy etc.), cross arms, stay sets, go switches etc. type of cross arms, etc.
- KU13.** tools and equipment used in testing, repair and maintenance Tools: e.g. plier, screwdriver, wrench set, hammer, drilling machine, hacksaw / cutting tools, measuring tape, pulleys (force pulley with sling), tommy bar, crimping machine, round / flat file, earth rod (discharge rod), leakage current monitoring kit
- KU14.** specific health and safety precautions which must be taken when carrying out t&d lines repair and maintenance work especially live line or equipment precautions: e.g. loose dhotis, pajamas, key chain or watch chains should not be worn; shoes with projecting nails or other types of metal parts not to be used; do not start work unless circuit is in off condition, line clear permit is taken on equipment, equipment or line is properly earthed, every electrical line or equipment should be first made off and take line clear permit before taking the work in hand
- KU15.** types and sizes of conductors and cables
- KU16.** causes of conductor damage causes: aeolian vibration, sway oscillation, galloping, unbalanced loading
- KU17.** importance of following good housekeeping and fire prevention procedures
- KU18.** importance of following job instructions and defined maintenance procedures
- KU19.** components of distribution lines line components: e.g. cross arm, insulator, line hardware, x-brace, armor rod, conductor, jumper, copper bond, arching horn, spacer, gang operated switch, drop out fuse, lightning arrester, etc..
- KU20.** importance of leaving the work area and equipment in a safe and clean condition on completion of the repair and maintenance activities
- KU21.** importance of reporting problems in a timely manner
- KU22.** principles and practices of electrical safety
- KU23.** standard procedures how to deal with electric shocks and electrocutions to rescue and minimize damage and harm
- KU24.** personal protective equipment (PPE) and clothing that must be worn during the inspection, repair and maintenance activity and from where can it be obtained; ppe: e.g. safety helmet, safety glove, safety shoe, climbing harness, lanyard and tool belt (when climbing), earth rod (discharge rod), zola, safety rope

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** listen and interpret information correctly regarding various job specifications, health and safety instructions, memos, etc. applicable to the job in english and/or local language
- GS2.** check and clarify task-related information
- GS3.** liaise with appropriate authorities using correct protocol
- GS4.** communicate with people in respectful form and manner in line with organizational protocol



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- GS5.** undertake basic Numerical computations and calculations numerical computations: addition, subtraction, multiplication, division,
- GS6.** identify various basic, compound and Solid shapes as per dimensions given Basic shapes: square, rectangle, triangle, circle, quadrilaterals Solid shapes: cube, rectangular prism, cylinder
- GS7.** use appropriate measuring techniques and units of measurement
- GS8.** use appropriate units and number systems to express degree of accuracy Units and number systems representing degree of accuracy: decimals places, significant figures
- GS9.** use metric systems of measurement
- GS10.** participate in on-the-job and other learning, training and development interventions and assessments
- GS11.** clarify task related information with appropriate personnel or technical adviser
- GS12.** seek to improve and modify own work practices
- GS13.** maintain current knowledge of application standards, legislation, codes of practice and product/process developments
- GS14.** identify problems with work planning, procedures, output and behavior and their implications
- GS15.** prioritize and plan for problem solving
- GS16.** communicate problems appropriately to others
- GS17.** identify sources of information and support for problem solving
- GS18.** seek assistance and support from other sources to solve problems
- GS19.** identify effective resolution techniques
- GS20.** select and apply resolution techniques
- GS21.** seek evidence for problem resolution
- GS22.** plan, prioritize and sequence work operations as per job requirements
- GS23.** organize and analyze information relevant to work
- GS24.** basic concepts of shop-floor work productivity including waste reduction, efficient material usage and optimization of time
- GS25.** undertake and express new ideas and initiatives to others
- GS26.** modify work plan to overcome unforeseen difficulties or developments that occur as work progresses
- GS27.** participate in improvement procedures including process, quality and
- GS28.** internal/external customer/supplier relationships
- GS29.** ones competencies in new and different situations and contexts to achieve more
- GS30.** exercise restraint while expressing dissent and during conflict situations
- GS31.** avoid and manage distractions to be disciplined at work
- GS32.** manage own time for achieving better results
- GS33.** work in a team in order to achieve better results
- GS34.** identify and clarify work roles within a team
- GS35.** communicate and cooperate with others in the team for better results



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Working safely</i>	5	19	-	-
PC1. work safely at all times, complying with health and safety legislation, regulations and other relevant guidelines	2	5	-	-
PC2. adhere to procedures or systems in place for health and safety, personal protective equipment (ppe) and other relevant safety regulations for electrical and related operations	2	5	-	-
PC3. work following laid down procedures and instructions	1	3	-	-
PC4. ensure that all tools, equipment, etc. Are in a safe and usable condition and are kept at secured location	-	3	-	-
PC5. ensure work area is clean and safe from hazards before and after the job is completed	-	3	-	-
<i>Handling tools and equipment for repair and maintenance</i>	10	21	-	-
PC6. Performance Criteria Description *	2	3	-	-
PC7. identify various types of circuits and its components correctly	2	2	-	-
PC8. identify accurately and acquire correct tools, equipment required for various aspects of repair and maintenance of distribution lines and components	2	3	-	-
PC9. ensure the tools and equipment are well maintained and handled, and approved for use	-	3	-	-
PC10. select and use tools and equipment such as chainsaws, pruning saws, manlifts and chippers to trim trees or its branches where required	-	3	-	-
PC11. identify hazards of trimming trees such as limits of approach, public safety and step and touch potential	2	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. clean and handle transmission and distribution line tools according to standard procedures	2	4	-	-
<i>Assist in repair and maintenance of Power distribution lines</i>	9	23	-	-
PC13. follow instruction, job standards and specifications accurately and safely while assisting linemen in performing off-line overhead and underground line maintenance procedure according to company procedures	2	5	-	-
PC14. hold, align, load and unload, poles and related equipment as per instructions and job standards and requirements accurately and safely while assisting in pole dismantling and re-setting procedure as per company procedures	2	4	-	-
PC15. carry out conductor stringing procedures, paving conductor on the ground along the pole under the guidance of lineman/supervisor as per instructions	2	4	-	-
PC16. re-tighten stay wires as per standard procedure, on instructions	1	3	-	-
PC17. tighten nuts and bolts securely for aligning lines and under guidance of lineman	-	3	-	-
PC18. hold, handover components, measure, unpack, join, align, load, unload equipment as per instructions, job standards and requirements accurately and safely while assisting linemen in repair of conductors by splicing, jointing, using armor rods, line guards and vibration dampers	2	4	-	-
<i>Post-repair and maintenance activities</i>	-	13	-	-
PC19. deal promptly and effectively with problems within control, and seek help and guidance from the relevant people for problems that cannot be resolved	-	4	-	-
PC20. leave the work area in a safe and tidy condition on completion of the repair and maintenance activities	-	3	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC21. refer unresolved job related problems to appropriate personnel for support	-	3	-	-
PC22. monitor the problem and keep the supervisor informed about progress or any delays in resolving the problem	-	3	-	-
NOS Total	24	76	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	PSS/N0101
NOS Name	Assisting in repair and maintenance of Power Distribution Lines and components
Sector	Power
Sub-Sector	Distribution
Occupation	Distribution System Erection, Commissioning, Operation & Maintenance
NSQF Level	2
Credits	TBD
Version	1.0
Last Reviewed Date	13/09/2017
Next Review Date	31/03/2022
NSQC Clearance Date	20/07/2015



Qualification Pack

PSS/N0102: Preparation and assistance for erection of Power Distribution Lines

Description

This unit covers the competencies required by technical helpers to assist in the erection of Power Distribution Lines. This includes working with the crew to install poles, dismantle poles and lay wiring, handling of tools and equipment for installation and carrying out necessary tasks in a safe, efficient and effective manner.

Scope

This unit/task covers the following:

- Working safely
- Prepare for erection of Power Distribution Lines
- Support and assist in Erection of Power Distribution Lines
- Post erection activities

Elements and Performance Criteria

Working safely

To be competent, the user/individual on the job must be able to:

- PC1.** work safely at all times, complying with health and safety legislation, regulations and other relevant guidelines
- PC2.** adhere to procedures or systems in place for health and safety, personal protective equipment (ppe) and other relevant safety regulations for electrical and related operations
- PC3.** work following laid down procedures and instructions
- PC4.** ensure that all tools and equipment are in a safe and usable condition and are kept at secured location
- PC5.** ensure work area is clean and safe from hazards before and after the job is completed
- PC6.** clarify and accurately understand job requirements for specific operations as per instructions given
- PC7.** identify equipment and tools required for installation work
- PC8.** perform loading and unloading operations for pole tower parts, tools and equipment in a safe and efficient manner
- PC9.** report hazards to senior crew members
- PC10.** perform excavation operations for pole setting, as per requirement and specifications mentioned by the lineman or supervisor, in a safe and efficient manner
- PC11.** perform pole erection procedures as per requirements and instructions, in a safe and efficient manner
- PC12.** set poles to proper depth, and align properly to the satisfaction of the supervisor
- PC13.** measure materials accurately for concrete mixing as per instructions
- PC14.** mix concrete to required texture and consistency as per instructions given



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- PC15.** carry out pit filling as per requirement, as correct procedures
- PC16.** perform pole dismantling procedure where required
- PC17.** perform post-installation procedures for ensuring clean and safe environment in the work and surrounding area
- PC18.** remove the waste components safely and correctly
- PC19.** deal promptly and effectively with problems within control, and seek help and guidance from the relevant people for problems that cannot be resolved
- PC20.** leave the work area in a safe and tidy condition on completion of the erection activities
- PC21.** refer unresolved job related problems to appropriate personnel for support
- PC22.** monitor the problem and keep the lineman or supervisor informed about progress or any delays in resolving the problem

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** relevant legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions
- KU2.** relevant health and safety requirements applicable in the work place
- KU3.** own job role and responsibilities and sources for information pertaining to employment terms, entitlements, job role and responsibilities
- KU4.** reporting structure, inter-dependent functions, lines and procedures in the work area
- KU5.** how to engage with specialists for support in order to resolve incidents and service requests
- KU6.** importance of working in clean and safe environment practices and procedures
- KU7.** relevant people and their responsibilities within the work area
- KU8.** escalation matrix and procedures for reporting work and employment related issues
- KU9.** common electricity terminology and correct interpretation of the same terminology: e.g. current, voltage, resistance, kilowatt (kw), kilowatt hour: (kwh)(unit of electric consumption)
- KU10.** specific terminology used in distribution line work terminology: e.g. pole, tower, load, load shedding, etc.
- KU11.** elements of the power system; elements: e.g. generation, transmission, distribution, metering, etc
- KU12.** different types of material and accessories used in power t&d materials and accessories: e.g. supports (poles-steel, cement, wooden), conductors (sizes, current carrying capacity), conductor accessories, binding tape, binding wire, p.g. clamp, t clamp etc., insulators (pin, disc, shackle, guy etc.), cross arms, stay sets, go switches etc. type of cross arms, etc.
- KU13.** Tools and equipment used in testing, repair and maintenance tools: e.g. plier, screwdriver, wrench set, hammer, drilling machine, hacksaw / cutting tools, measuring tape, pulleys (force pulley with sling), tommy bar, crimping machine, round / flat file, earth rod (discharge rod), leakage current monitoring kit
- KU14.** tower parts and accessories



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- KU15.** Parts and accessories: e.g. insulator, machine bolts, suspension clamps, strain clamp, overhead earth wires, cross-arms and braces, conductors and accessories, opgw (optical ground wire), tower parts (lattices / members, step bolts, bolts and nuts, plates and back plates, grounding cables)
- KU16.** specific health and safety precautions which must be taken when carrying out pole erection procedures Safety requirements: e.g. poles securely fastened, warning devices are installed
- KU17.** Hazards associated with carrying out pole erection processes and how they can be minimized hazards: e.g. blockages and obstructions, live wires and equipment, unsecured ladders, etc.
- KU18.** importance of following job instructions and defined procedures for tower/pole erection
- KU19.** importance of leaving the work area and equipment in a safe and clean condition on completion of the erection activities
- KU20.** importance of reporting problems in a timely manner
- KU21.** importance of tools and equipment to be kept in a safe and usable condition
- KU22.** personal protective equipment (ppe) and clothing that must be worn during the heat treatment activity and from where can it be obtained

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** listen and interpret information correctly regarding various job specifications, health and safety instructions, memos, etc. applicable to the job in english and/or local language
- GS2.** convey and share technical information clearly using appropriate language
- GS3.** check and clarify task-related information
- GS4.** liaise with appropriate authorities using correct protocol
- GS5.** communicate with people in respectful form and manner in line with organizational protocol
- GS6.** undertake basic numerical computations and calculations Numerical computations: addition, subtraction, multiplication, division,
- GS7.** identify various basic, compound and solid shapes as per dimensions given Basic shapes: square, rectangle, triangle, circle, quadrilaterals Solid shapes: cube, rectangular prism, cylinder
- GS8.** use appropriate measuring techniques and units of measurement
- GS9.** use appropriate units and number systems to express degree of accuracy Units and number systems representing degree of accuracy: decimals places, significant figures
- GS10.** use metric systems of measurement
- GS11.** participate in on-the-job and other learning, training and development interventions and assessments
- GS12.** clarify task related information with appropriate personnel or technical adviser SA13. seek to
- GS13.** seek to improve and modify own work practices
- GS14.** seek to improve and modify own work practices
- GS15.** identify problems with work planning, procedures, output and behavior and their implications
- GS16.** prioritize and plan for problem solving
- GS17.** communicate problems appropriately to others



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- GS18.** identify sources of information and support for problem solving
- GS19.** seek assistance and support from other sources to solve problems
- GS20.** identify effective resolution techniques
- GS21.** select and apply resolution techniques
- GS22.** seek evidence for problem resolution
- GS23.** plan, prioritize and sequence work operations as per job requirements
- GS24.** organize and analyze information relevant to work
- GS25.** basic concepts of shop-floor work productivity including waste reduction, efficient material usage and optimization of time
- GS26.** undertake and express new ideas and initiatives to others
- GS27.** modify work plan to overcome unforeseen difficulties or developments that occur as work progresses
- GS28.** participate in improvement procedures including process, quality and internal/external customer/supplier relationships
- GS29.** ones competencies in new and different situations and contexts to achieve more
- GS30.** exercise restraint while expressing dissent and during conflict situations
- GS31.** avoid and manage distractions to be disciplined at work
- GS32.** manage own time for achieving better results
- GS33.** work in a team in order to achieve better results
- GS34.** identify and clarify work roles within a team
- GS35.** communicate and cooperate with others in the team for better results
- GS36.** communicate and cooperate with others in the team for better results

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Working safely</i>	25	75	-	-
PC1. work safely at all times, complying with health and safety legislation, regulations and other relevant guidelines	2	4	-	-
PC2. adhere to procedures or systems in place for health and safety, personal protective equipment (ppe) and other relevant safety regulations for electrical and related operations	2	4	-	-
PC3. work following laid down procedures and instructions	1	3	-	-
PC4. ensure that all tools and equipment are in a safe and usable condition and are kept at secured location	-	3	-	-
PC5. ensure work area is clean and safe from hazards before and after the job is completed	1	3	-	-
PC6. clarify and accurately understand job requirements for specific operations as per instructions given	1	3	-	-
PC7. identify equipment and tools required for installation work	1	3	-	-
PC8. perform loading and unloading operations for pole tower parts, tools and equipment in a safe and efficient manner	1	3	-	-
PC9. report hazards to senior crew members	1	3	-	-
PC10. perform excavation operations for pole setting, as per requirement and specifications mentioned by the lineman or supervisor, in a safe and efficient manner	2	4	-	-
PC11. perform pole erection procedures as per requirements and instructions, in a safe and efficient manner	2	4	-	-
PC12. set poles to proper depth, and align properly to the satisfaction of the supervisor	1	4	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. measure materials accurately for concrete mixing as per instructions	1	4	-	-
PC14. mix concrete to required texture and consistency as per instructions given	1	3	-	-
PC15. carry out pit filling as per requirement, as correct procedures	1	4	-	-
PC16. perform pole dismantling procedure where required	1	4	-	-
PC17. perform post-installation procedures for ensuring clean and safe environment in the work and surrounding area	1	4	-	-
PC18. remove the waste components safely and correctly	1	3	-	-
PC19. deal promptly and effectively with problems within control, and seek help and guidance from the relevant people for problems that cannot be resolved	1	3	-	-
PC20. leave the work area in a safe and tidy condition on completion of the erection activities	1	2	-	-
PC21. refer unresolved job related problems to appropriate personnel for support	1	3	-	-
PC22. monitor the problem and keep the lineman or supervisor informed about progress or any delays in resolving the problem	1	4	-	-
NOS Total	25	75	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	PSS/N0102
NOS Name	Preparation and assistance for erection of Power Distribution Lines
Sector	Power
Sub-Sector	Distribution
Occupation	Distribution System Erection, Commissioning, Operation & Maintenance
NSQF Level	2
Credits	TBD
Version	1.0
Last Reviewed Date	13/09/2017
Next Review Date	31/03/2022
NSQC Clearance Date	20/07/2015



Qualification Pack

PSS/N0103: Assisting in operation and maintenance of 11/0.433 KV Distribution Substation

Description

This unit covers the competencies required by technical helpers to support and assist in erection and maintenance for an 11/0.433 KV Distribution Substation. This includes working with the crew to install the Substation transformer, handling of tools and equipment for installation and maintenance and carrying out necessary tasks in a safe, efficient and effective manner.

Scope

This unit/task covers the following:

- Working safely
- Assisting to erect an 11/0.433 KV Distribution Substation
- Assisting in carrying out maintenance for the Distribution Substation
- Post erection activities

Elements and Performance Criteria

Working safely

To be competent, the user/individual on the job must be able to:

- PC1.** work safely at all times, complying with health and safety legislation, regulations and other relevant guidelines
- PC2.** adhere to procedures or systems in place for health and safety, personal protective equipment (PPE) and other relevant safety regulations for electrical and related operations
- PC3.** work following laid down procedures and instructions
- PC4.** ensure that all tools, equipment, power cables are in a safe and usable condition and are kept at secured location
- PC5.** ensure work area is clean and safe from hazards before and after the job is completed

Assisting to erect an 11/0.433 KV Distribution Substation

To be competent, the user/individual on the job must be able to:

- PC6.** clarify and understand accurately job requirements for specific operations as per instructions given by supervisor
- PC7.** identify various components of the power system
- PC8.** identify equipment and tools required for installation work accurately
- PC9.** inspect the component to check if it is as per specification and without defects
- PC10.** identify, measure and acquire correct materials required for the substation erection and installation work
- PC11.** perform loading and unloading operations for substation parts in a safe and efficient manner
- PC12.** assist in lifting the transformer and putting it on the transformer bed in a safe and efficient manner

Assisting in carrying out maintenance of the substation



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To be competent, the user/individual on the job must be able to:

- PC13.** hand over correct equipment and tools to supervisors and senior technicians in a safe manner
- PC14.** clean bushings, transformer and other components safely and as per correct procedure
- PC15.** remove dust & dirt deposition, salt or chemical deposition, cement or acid fumes depositions
- PC16.** clean conservator from inside as per instruction and standard procedure
- PC17.** climb ladders and elevated platforms and structures to carry out maintenance and other activities, safely

Post erection activities

To be competent, the user/individual on the job must be able to:

- PC18.** layout gravel in the substation as required
- PC19.** assist in fixing the fence to secure the substation in a safe, efficient manner as per correct procedure
- PC20.** fix rating plate and diagram plate as per requirement
- PC21.** deal promptly and effectively with problems within control, and seek help and guidance from the relevant people for problems that cannot be resolved
- PC22.** leave the work area in a safe and tidy condition on completion of the substation construction and maintenance activities
- PC23.** refer unresolved job related problems to appropriate personnel for support

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** relevant legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions
- KU2.** relevant health and safety requirements applicable in the work place
- KU3.** own job role and responsibilities and sources for information pertaining to employment terms, entitlements, job role and responsibilities
- KU4.** reporting structure, inter-dependent functions, lines and procedures in the work area
- KU5.** how to engage with specialists for support in order to resolve incidents and service requests
- KU6.** importance of working in clean and safe environment practices and procedures
- KU7.** relevant people and their responsibilities within the work area
- KU8.** escalation matrix and procedures for reporting work and employment related issues
- KU9.** various components of the power system Components: e.g. transformers
- KU10.** specific health and safety precautions which must be taken when carrying out substation installation and maintenance processes
- KU11.** Hazards associated with carrying out substation construction and installation process and maintenance, and how they can be minimized hazards: e.g. live wires and equipment, heavy objects, insects and reptiles, obstructions and blockages, sharp edges and equipment, etc.
- KU12.** importance of following job instructions and defined installation and maintenance procedures
- KU13.** equipment used in substation construction and maintenance activities



Qualification Pack

- KU14.** importance of leaving the work area and equipment in a safe and clean condition on completion of the heat treatment activities
- KU15.** importance of reporting problems in a timely manner
- KU16.** importance of tools and equipment to be kept in a safe and usable condition
- KU17.** importance of using gravel in paving substation surfaces
- KU18.** importance of displaying rating and diagram plates
- KU19.** personal protective equipment (ppe) and clothing that must be worn during the heat treatment activity and from where can it be obtained

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** listen and interpret information correctly regarding various job specifications, health and safety instructions, memos, etc. applicable to the job in english and/or local language
- GS2.** convey and share technical information clearly using appropriate language
- GS3.** check and clarify task-related information
- GS4.** liaise with appropriate authorities using correct protocol
- GS5.** communicate with people in respectful form and manner in line with organizational protocol
- GS6.** undertake basic Numerical computations and calculations numerical computations: addition, subtraction, multiplication, division,
- GS7.** identify various basic, compound and solid shapes as per dimensions given Basic shapes: square, rectangle, triangle, circle, quadrilaterals Solid shapes: cube, rectangular prism, cylinder
- GS8.** use appropriate measuring techniques and units of measurement
- GS9.** use appropriate units and number systems to express degree of accuracy Units and number systems representing degree of accuracy: decimals places, significant figures
- GS10.** use metric systems of measurement
- GS11.** participate in on-the-job and other learning, training and development interventions and assessments
- GS12.** clarify task related information with appropriate personnel or technical adviser
- GS13.** seek to improve and modify own work practices
- GS14.** maintain current knowledge of application standards, legislation, codes of practice and product/process developments
- GS15.** identify problems with work planning, procedures, output and behavior and their implications
- GS16.** prioritize and plan for problem solving
- GS17.** communicate problems appropriately to others
- GS18.** identify sources of information and support for problem solving
- GS19.** seek assistance and support from other sources to solve problems
- GS20.** identify effective resolution techniques
- GS21.** select and apply resolution techniques
- GS22.** seek evidence for problem resolution
- GS23.** plan, prioritize and sequence work operations as per job requirements



Qualification Pack

- GS24.** organize and analyze information relevant to work
- GS25.** basic concepts of shop-floor work productivity including waste reduction, efficient material usage and optimization of time
- GS26.** undertake and express new ideas and initiatives to others
- GS27.** modify work plan to overcome unforeseen difficulties or developments that occur as work progresses
- GS28.** participate in improvement procedures including process, quality and internal/external customer/supplier relationships
- GS29.** ones competencies in new and different situations and contexts to achieve more
- GS30.** exercise restraint while expressing dissent and during conflict situations
- GS31.** avoid and manage distractions to be disciplined at work
- GS32.** manage own time for achieving better results
- GS33.** work in a team in order to achieve better results
- GS34.** identify and clarify work roles within a team
- GS35.** communicate and cooperate with others in the team for better results
- GS36.** seek assistance from fellow team members

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Working safely</i>	5	17	-	-
PC1. work safely at all times, complying with health and safety legislation, regulations and other relevant guidelines	2	4	-	-
PC2. adhere to procedures or systems in place for health and safety, personal protective equipment (PPE) and other relevant safety regulations for electrical and related operations	2	4	-	-
PC3. work following laid down procedures and instructions	1	3	-	-
PC4. ensure that all tools, equipment, power cables are in a safe and usable condition and are kept at secured location	-	3	-	-
PC5. ensure work area is clean and safe from hazards before and after the job is completed	-	3	-	-
<i>Assisting to erect an 11/0.433 KVDistribution Substation</i>	10	23	-	-
PC6. clarify and understand accurately job requirements for specific operations as per instructions given by supervisor	2	3	-	-
PC7. identify various components of the power system	2	2	-	-
PC8. identify equipment and tools required for installation work accurately	2	2	-	-
PC9. inspect the component to check if it is as per specification and without defects	1	4	-	-
PC10. identify, measure and acquire correct materials required for the substation erection and installation work	1	5	-	-
PC11. perform loading and unloading operations for substation parts in a safe and efficient manner	-	3	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. assist in lifting the transformer and putting it on the transformer bed in a safe and efficient manner	2	4	-	-
<i>Assisting in carrying out maintenance of the substation</i>	5	16	-	-
PC13. hand over correct equipment and tools to supervisors and senior technicians in a safe manner	1	2	-	-
PC14. clean bushings, transformer and other components safely and as per correct procedure	1	4	-	-
PC15. remove dust & dirt deposition, salt or chemical deposition, cement or acid fumes depositions	-	3	-	-
PC16. clean conservator from inside as per instruction and standard procedure	2	3	-	-
PC17. climb ladders and elevated platforms and structures to carry out maintenance and other activities, safely	1	4	-	-
<i>Post erection activities</i>	5	19	-	-
PC18. layout gravel in the substation as required	1	2	-	-
PC19. assist in fixing the fence to secure the substation in a safe, efficient manner as per correct procedure	2	4	-	-
PC20. fix rating plate and diagram plate as per requirement	2	4	-	-
PC21. deal promptly and effectively with problems within control, and seek help and guidance from the relevant people for problems that cannot be resolved	-	3	-	-
PC22. leave the work area in a safe and tidy condition on completion of the substation construction and maintenance activities	-	3	-	-
PC23. refer unresolved job related problems to appropriate personnel for support	-	3	-	-
NOS Total	25	75	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N0103
NOS Name	Assisting in operation and maintenance of 11/0.433 KV Distribution Substation
Sector	Power
Sub-Sector	Distribution
Occupation	Distribution System Erection, Commissioning, Operation & Maintenance
NSQF Level	2
Credits	TBD
Version	1.0
Last Reviewed Date	13/09/2017
Next Review Date	31/03/2022
NSQC Clearance Date	20/07/2015



Qualification Pack

PSS/N0104: Assisting in laying of underground and AB cables

Description

This unit covers the competencies required by technical helpers to support linemen in laying underground and AB cables for setting up Power Distribution Lines. This includes digging trenches, prepare and lay wiring, handling of tools and equipment for laying cables and carrying out necessary tasks in a safe, efficient and effective manner. The candidate will be expected to perform under complete supervision and as per instructions given.

Scope

This unit/task covers the following:

- Working safely
- Preparing for laying cables
- Assisting in laying and maintenance of cables
- Post laying activities

Elements and Performance Criteria

Working safely

To be competent, the user/individual on the job must be able to:

- PC1.** work safely at all times, complying with health and safety legislation, regulations and other relevant guidelines
- PC2.** adhere to procedures or systems in place for health and safety, personal protective equipment (PPE) and other relevant safety regulations for electrical and related operations
- PC3.** work following laid down procedures and instructions
- PC4.** ensure that all tools, equipment, power cables are in a safe and usable condition and are kept at secured location
- PC5.** ensure work area is clean and safe from hazards before and after the job is completed

Preparing materials for job

To be competent, the user/individual on the job must be able to:

- PC6.** clarify job requirements for specific operations as per instructions given from the supervisor
- PC7.** identify all tools and equipment required for the work are acquired and transported safely to the work site
- PC8.** perform loading and unloading of materials, tools and equipment in a safe and efficient manner

Assisting in laying and maintenance of cables

To be competent, the user/individual on the job must be able to:

- PC9.** perform trench digging operations as per requirement and specifications, in a safe and efficient manner
- PC10.** pull cable through conduit using equipment such as tension machines, winches and capstans
- PC11.** measure materials required for the concrete mix and mixing them in proportions specified by the supervisor



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Post laying operations

To be competent, the user/individual on the job must be able to:

- PC12.** deal promptly and effectively with problems within control, and seek help and guidance from the relevant people for problems that cannot be resolved
- PC13.** perform post-installation procedures for ensuring clean and safe environment in the work and surrounding area
- PC14.** refer unresolved job related problems to appropriate personnel for support
- PC15.** monitor the problem and keep the lineman or supervisor informed about progress or any delays in resolving the problem
- PC16.** transport and store materials, equipment, tools as per laid down practice, in as safe and efficient manner

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** relevant legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions
- KU2.** relevant health and safety requirements applicable in the work place
- KU3.** own job role and responsibilities and sources for information pertaining to employment terms, entitlements, job role and responsibilities
- KU4.** reporting structure, inter-dependent functions, lines and procedures in the work area
- KU5.** how to engage with specialists for support in order to resolve incidents and service requests
- KU6.** importance of working in clean and safe environment practices and procedures
- KU7.** relevant people and their responsibilities within the work area
- KU8.** escalation matrix and procedures for reporting work and employment related issues
- KU9.** common electric and power terminology used in distribution
- KU10.** specific health and safety precautions which must be taken when carrying out cable laying processes and working in confined spaces. Precautions: e.g. loose dhotis, pajamas, key chain or watch chains should not be worn; shoes with projecting nails or other types of metal parts not to be used; do not start work unless circuit is in off condition, line clear permit is taken on equipment, equipment or line is properly earthed, every electrical line or equipment should be first made off and take line clear permit before taking the work in hand
- KU11.** hazards associated with carrying out cable laying processes and how they can be minimized. Hazards: live wires and equipment, blockages and obstructions, loose earth, sharp surfaces and edges, insects and reptiles, heavy objects, etc.
- KU12.** importance of following job instructions and defined cable laying procedures
- KU13.** preparation of cables and equipment for cable laying activities
- KU14.** adjacent utilities such as gas, water, communication and drainage requirements
- KU15.** tools and equipment used in cable laying activities
- KU16.** trench digging techniques and related safety measures
- KU17.** concrete mixing techniques and materials involved
- KU18.** back filling procedures
- KU19.** pulling methods and calculations



Qualification Pack

- KU20.** importance of leaving the work area and equipment in a safe and clean condition on completion of the job activities
- KU21.** importance of reporting problems in a timely manner
- KU22.** importance of tools and equipment to be kept in a safe and usable condition
- KU23.** personal protective equipment (PPE) and clothing that must be worn during the cable laying and maintenance activity and from where can it be obtained ppe: e.g. safety helmet, safety glove, safety shoe, climbing harness, lanyard and tool belt (when climbing), earth rod (discharge rod), zola, safety rope

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** listen and interpret information correctly regarding various job specifications, health and safety instructions, memos, etc. applicable to the job in English and/or local language
- GS2.** convey and share technical information clearly using appropriate language
- GS3.** check and clarify task-related information
- GS4.** liaise with appropriate authorities using correct protocol
- GS5.** communicate with people in respectful form and manner in line with organizational protocol
- GS6.** undertake basic Numerical computations and calculations numerical computations: addition, subtraction, multiplication, division,
- GS7.** identify various basic, compound and solid shapes as per dimensions given Basic shapes: square, rectangle, triangle, circle, quadrilaterals Solid shapes: cube, rectangular prism, cylinder
- GS8.** use appropriate measuring techniques and units of measurement
- GS9.** use appropriate units and number systems to express degree of accuracy Units and number systems representing degree of accuracy: decimals places, significant figures
- GS10.** use metric systems of measurement
- GS11.** participate in on-the-job and other learning, training and development interventions and assessments
- GS12.** clarify task related information with appropriate personnel or technical adviser
- GS13.** seek to improve and modify own work practices
- GS14.** maintain current knowledge of application standards, legislation, codes of practice and product/process developments
- GS15.** identify problems with work planning, procedures, output and behavior and their implications
- GS16.** prioritize and plan for problem solving
- GS17.** communicate problems appropriately to others
- GS18.** identify sources of information and support for problem solving
- GS19.** seek assistance and support from other sources to solve problems
- GS20.** identify effective resolution techniques
- GS21.** select and apply resolution techniques
- GS22.** seek evidence for problem resolution
- GS23.** plan, prioritize and sequence work operations as per job requirements



Qualification Pack

- GS24.** organize and analyze information relevant to work
- GS25.** basic concepts of shop-floor work productivity including waste reduction, efficient material usage and optimization of time
- GS26.** undertake and express new ideas and initiatives to others
- GS27.** modify work plan to overcome unforeseen difficulties or developments that occur as work progresses
- GS28.** participate in improvement procedures including process, quality and internal/external customer/supplier relationships
- GS29.** ones competencies in new and different situations and contexts to achieve more
- GS30.** exercise restraint while expressing dissent and during conflict situations
- GS31.** avoid and manage distractions to be disciplined at work
- GS32.** manage own time for achieving better results
- GS33.** work in a team in order to achieve better results
- GS34.** identify and clarify work roles within a team
- GS35.** communicate and cooperate with others in the team for better results
- GS36.** seek assistance from fellow team members

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Working safely</i>	7	22	-	-
PC1. work safely at all times, complying with health and safety legislation, regulations and other relevant guidelines	2	5	-	-
PC2. adhere to procedures or systems in place for health and safety, personal protective equipment (PPE) and other relevant safety regulations for electrical and related operations	2	5	-	-
PC3. work following laid down procedures and instructions	1	4	-	-
PC4. ensure that all tools, equipment, power cables are in a safe and usable condition and are kept at secured location	1	4	-	-
PC5. ensure work area is clean and safe from hazards before and after the job is completed	1	4	-	-
<i>Preparing materials for job</i>	6	13	-	-
PC6. clarify job requirements for specific operations as per instructions given from the supervisor	2	4	-	-
PC7. identify all tools and equipment required for the work are acquired and transported safely to the work site	2	4	-	-
PC8. perform loading and unloading of materials, tools and equipment in a safe and efficient manner	2	5	-	-
<i>Assisting in laying and maintenance of cables</i>	7	15	-	-
PC9. perform trench digging operations as per requirement and specifications, in a safe and efficient manner	2	6	-	-
PC10. pull cable through conduit using equipment such as tension machines, winches and capstans	2	4	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. measure materials required for the concrete mix and mixing them in proportions specified by the supervisor	3	5	-	-
<i>Post laying operations</i>	5	25	-	-
PC12. deal promptly and effectively with problems within control, and seek help and guidance from the relevant people for problems that cannot be resolved	-	5	-	-
PC13. perform post-installation procedures for ensuring clean and safe environment in the work and surrounding area	2	5	-	-
PC14. refer unresolved job related problems to appropriate personnel for support	-	5	-	-
PC15. monitor the problem and keep the lineman or supervisor informed about progress or any delays in resolving the problem	1	5	-	-
PC16. transport and store materials, equipment, tools as per laid down practice, in as safe and efficient manner	2	5	-	-
NOS Total	25	75	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N0104
NOS Name	Assisting in laying of underground and AB cables
Sector	Power
Sub-Sector	Distribution
Occupation	Distribution System Erection, Commissioning, Operation & Maintenance
NSQF Level	2
Credits	TBD
Version	1.0
Last Reviewed Date	13/09/2017
Next Review Date	31/03/2022
NSQC Clearance Date	20/07/2015



Qualification Pack

PSS/N1336: Working effectively with others

Description

This unit covers basic etiquette and competencies that a candidate is required to possess and demonstrate in their behavior and interactions with others at the workplace. These cover areas such as communication etiquette, discipline, listening, handling conflict and grievances.

Elements and Performance Criteria

working with others

To be competent, the user/individual on the job must be able to:

- PC1.** accurately receive information and instructions from the supervisor and fellow workers, getting clarification where required
- PC2.** accurately pass on information to authorized persons who require it and within agreed timescale and confirm its receipt
- PC3.** give information to others clearly, at a pace and in a manner that helps them to understand
- PC4.** display helpful behavior by assisting others in performing tasks in a positive manner, where required and possible
- PC5.** consult with and assist others to maximize effectiveness and efficiency in carrying out tasks
- PC6.** display appropriate communication etiquette while working
- PC7.** display active listening skills while interacting with others at work
- PC8.** use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism
- PC9.** demonstrate responsible and disciplined behaviors at the workplace
- PC10.** escalate grievances and problems to appropriate authority as per procedure to resolve them and avoid conflict

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions
- KU2.** reporting structure, inter-dependent functions, lines and procedures in the work area
- KU3.** relevant people and their responsibilities within the work area
- KU4.** escalation matrix and procedures for reporting work and employment related issues
- KU5.** various categories of people that one is required to communicate and co-ordinate with in the organization
- KU6.** importance of effective communication in the workplace
- KU7.** importance of teamwork in organizational and individual success
- KU8.** various components of effective communication
- KU9.** key elements of active listening
- KU10.** value and importance of active listening and assertive communication



Qualification Pack

- KU11.** barriers to effective communication
- KU12.** importance of tone and pitch in effective communication
- KU13.** importance of avoiding casual expletives and unpleasant terms while communicating professional circles
- KU14.** how poor communication practices can disturb people, environment and cause problems for the employee, the employer and the customer
- KU15.** importance of ethics for professional success
- KU16.** importance of discipline for professional success
- KU17.** what constitutes disciplined behavior for a working professional
- KU18.** common reasons for interpersonal conflict
- KU19.** importance of developing effective working relationships for professional success
- KU20.** expressing and addressing grievances appropriately and effectively
- KU21.** importance and ways of managing interpersonal conflict effectively

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** note the information communicated by the officer incharge
- GS2.** note down observations (if any) related to the operation/maintenance
- GS3.** read and interpret the process required for different types of manuals
- GS4.** read and interpret the flowchart of all parts of an assembly
- GS5.** read manuals and documents to understand the product-details & how they can be used
- GS6.** discuss task lists, schedules and activities with the colleague/supervisor
- GS7.** effectively communicate with the team members
- GS8.** attentively listen and comprehend the information given by the colleague/supervisor/contractor
- GS9.** communicate clearly with the colleague on the issues faced during query/fault
- GS10.** follow colleague/contractor rule-based decision making process
- GS11.** take decisions with systematic course of actions and/or response
- GS12.** planning and organization of tasks to meet deadlines
- GS13.** build customer relationships and use customer centric approach
- GS14.** seek and comprehend operation related inputs for clarification find ways of modifying difficult operating stages to make it operation friendly
- GS15.** work systematically and logically to resolve the issues and identify causation and anticipate unexpected results. Quick approach and solution towards faults repairing
- GS16.** critically evaluate operation parameters in relation to system normality develop a holistic and comprehensive profile of grid station on segregated discrete processes

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>working with others</i>	30	70	-	-
PC1. accurately receive information and instructions from the supervisor and fellow workers, getting clarification where required	3	7	-	-
PC2. accurately pass on information to authorized persons who require it and within agreed timescale and confirm its receipt	3	7	-	-
PC3. give information to others clearly, at a pace and in a manner that helps them to understand	3	7	-	-
PC4. display helpful behavior by assisting others in performing tasks in a positive manner, where required and possible	3	7	-	-
PC5. consult with and assist others to maximize effectiveness and efficiency in carrying out tasks	3	7	-	-
PC6. display appropriate communication etiquette while working	3	7	-	-
PC7. display active listening skills while interacting with others at work	3	7	-	-
PC8. use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism	3	7	-	-
PC9. demonstrate responsible and disciplined behaviors at the workplace	3	7	-	-
PC10. escalate grievances and problems to appropriate authority as per procedure to resolve them and avoid conflict	3	7	-	-
NOS Total	30	70	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N1336
NOS Name	Working effectively with others
Sector	Power
Sub-Sector	Generic
Occupation	Generic
NSQF Level	2
Credits	TBD
Version	1.0
Last Reviewed Date	31/03/2022
Next Review Date	31/03/2025
NSQC Clearance Date	01/03/2023



Qualification Pack

PSS/N2001: Use basic health and safety practices for power related work

Description

This unit covers health, safety and security for power related work. This includes procedures and practices that candidates need to follow to help maintain a healthy, safe and secure work environment in a power plant, power station/substation or on the field while working on power equipment. It covers responsibilities towards self, others, assets and the environment

Elements and Performance Criteria

Health and safety

To be competent, the user/individual on the job must be able to:

- PC1.** use protective clothing/equipment for specific tasks and work conditions
- PC2.** state the name and location of people responsible for health and safety in the workplace
- PC3.** state the names and location of documents that refer to health and safety in the workplace
- PC4.** identify job-site hazardous work and state possible causes of risk or accident in the workplace
- PC5.** follow electrical safe working procedures such as tag out/lock out and display ptw (permit to work),
- PC6.** follow warning signs (danger, out of service, etc.) while working with electrical systems
- PC7.** use standard safe working practices when working at heights, confined areas and trenches
- PC8.** test any electrical equipment and system using insulated testing devices before touching them
- PC9.** ensure positive isolation of electrical equipment & system as per given standards
- PC10.** recognize any abnormalities in electrical equipment or system installed alarm annunciation and/or noticing parameters from gauge/ indicator installed
- PC11.** carry out safe working practices while dealing with hazards to ensure the safety of self and others
- PC12.** state methods of accident prevention in the work environment of the job role
- PC13.** state location of general health and safety equipment in the workplace
- PC14.** inspect for faults, set up and safely use of scaffolds and elevated platforms and ladder
- PC15.** lift, carry and transport heavy objects & tools safely using correct procedures from storage to workplace and vice versa
- PC16.** inspect grid station and its equipment routinely for any signs of oil and water leakage
- PC17.** store flammable materials and machine lubricating oil safely and correctly
- PC18.** check that the emission and pollution control devices are working properly in line with environmental policy standards
- PC19.** ensure proper working condition of battery and battery charger
- PC20.** maintain electrolyte level of each cell with distilled water
- PC21.** maintain proper ventilation in battery room
- PC22.** apply good housekeeping practices at all times
- PC23.** identify common hazard signs displayed in various areas



Qualification Pack

PC24. retrieve and/or point out documents that refer to health and safety in the workplace

PC25. inform relevant authorities about any abnormal situation/behavior of any equipment/system promptly

Fire Safety

To be competent, the user/individual on the job must be able to:

PC26. use the various appropriate fire extinguishers on different types of fires correctly

PC27. distinguish between various types of fire

PC28. demonstrate rescue techniques applied during fire hazard

PC29. demonstrate good housekeeping in order to prevent fire hazards

PC30. demonstrate the correct use of a fire extinguisher

Emergencies, rescue and first-aid procedures

To be competent, the user/individual on the job must be able to:

PC31. demonstrate how to free a person from electrocution

PC32. administer appropriate first aid to victims where required e.g. in case of bleeding, burns, choking, electric shock, poisoning etc.

PC33. demonstrate basic techniques of bandaging

PC34. respond promptly and appropriately to an accident situation or medical emergency in real or simulated environments

PC35. perform and organize loss minimization or rescue activity during an accident in real or simulated environments

PC36. administer first aid to victims in case of a heart attack or cardiac arrest due to electric shock, before the arrival of emergency services in real or simulated cases

PC37. demonstrate the artificial respiration and the cpr process

PC38. participate in emergency procedures emergency procedures: raising alarm, safe/efficient, evacuation, correct means of escape, correct assembly point, roll call, correct return to work

PC39. write accident/incident report or dictate a report to another person, and send report to person responsible

PC40. demonstrate correct method to move injured people and others during an emergency

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions

KU2. reporting structure, inter-dependent functions, lines and procedures in the work area

KU3. relevant people and their responsibilities within the work area

KU4. escalation matrix and procedures for reporting work and employment related issues

KU5. meaning of hazards and risks

KU6. health and safety hazards commonly present in the work environment and related precautions

KU7. possible causes of risk, hazard or accident in the workplace and why risk and/or accidents are possible



Qualification Pack

- KU8.** possible causes of risk and accident
- KU9.** methods of accident prevention
- KU10.** safe working practices when working with tools and machines
- KU11.** safe working practices while working at various hazardous sites
- KU12.** where to find all the general health and safety equipment in the workplace
- KU13.** various dangers associated with the use of electrical equipment
- KU14.** positive isolation of electrical equipment and system
- KU15.** safe handling and disposal of hazardous power plant wastes
- KU16.** use of emission and pollution control devices and measures taken to control pollution
- KU17.** various safety procedures and equipment used to work at heights, trenches and confined places
- KU18.** safe working practices specific to working with electrical equipment & system e.g. lock out/tag out, ptw, etc.
- KU19.** preventative and remedial actions to be taken in the case of exposure to toxic materials
- KU20.** importance of using protective clothing/equipment and other insulated work gear while handling electrical system and equipment
- KU21.** precautionary activities taken to prevent fire accident
- KU22.** various causes of fire
- KU23.** techniques of using the different fire extinguishers
- KU24.** different methods of extinguishing fire
- KU25.** different materials used for extinguishing fire
- KU26.** emergency rescue techniques applied during a fire hazard
- KU27.** various types of safety signs and what they mean
- KU28.** appropriate basic first aid treatment relevant to the condition e.g. shock, electrical shock, bleeding, breaks to bones, minor burns, resuscitation, poisoning, eye injuries

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** note the information communicated by the officer incharge.
- GS2.** note down observations (if any) related to the operation/maintenance.
- GS3.** read and interpret the process required for different types of manuals for maintenance.
- GS4.** read and interpret the flowchart of all parts of an assembly.
- GS5.** read manuals and documents to understand the product-details & how they can be used.
- GS6.** discuss task lists, schedules and activities with the colleague/supervisor.
- GS7.** effectively communicate with the team members.
- GS8.** attentively listen and comprehend the information given by the colleague/supervisor/contractor.
- GS9.** communicate clearly with the colleague on the issues faced during query/fault.
- GS10.** follow colleague/contractor rule-based decision making process.
- GS11.** take decisions with systematic course of actions and/or response.



Qualification Pack

- GS12.** planning and organization of tasks to meet deadlines.
- GS13.** build customer relationships and use customer centric approach.
- GS14.** seek and comprehend operation related inputs for clarification
- GS15.** find ways of modifying difficult operating stages to make it operation friendly
- GS16.** work systematically and logically to resolve the issues and identify causation and anticipate unexpected results.
- GS17.** quick approach and solution towards faults repairing.
- GS18.** critically evaluate operation parameters in relation to system normality
- GS19.** develop a holistic and comprehensive profile of grid station on segregated discrete process.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Health and safety</i>	11	46	-	-
PC1. use protective clothing/equipment for specific tasks and work conditions	-	3	-	-
PC2. state the name and location of people responsible for health and safety in the workplace	-	2	-	-
PC3. state the names and location of documents that refer to health and safety in the workplace	-	2	-	-
PC4. identify job-site hazardous work and state possible causes of risk or accident in the workplace	1	1	-	-
PC5. follow electrical safe working procedures such as tag out/lock out and display ptw (permit to work),	1	2	-	-
PC6. follow warning signs (danger, out of service, etc.) while working with electrical systems	1	1	-	-
PC7. use standard safe working practices when working at heights, confined areas and trenches	1	2	-	-
PC8. test any electrical equipment and system using insulated testing devices before touching them	1	2	-	-
PC9. ensure positive isolation of electrical equipment & system as per given standards	1	2	-	-
PC10. recognize any abnormalities in electrical equipment or system installed alarm annunciation and/or noticing parameters from gauge/ indicator installed	1	2	-	-
PC11. carry out safe working practices while dealing with hazards to ensure the safety of self and others	1	1	-	-
PC12. state methods of accident prevention in the work environment of the job role	-	2	-	-
PC13. state location of general health and safety equipment in the workplace	-	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. inspect for faults, set up and safely use of scaffolds and elevated platforms and ladder	-	2	-	-
PC15. lift, carry and transport heavy objects & tools safely using correct procedures from storage to workplace and vice versa	-	2	-	-
PC16. inspect grid station and its equipment routinely for any signs of oil and water leakage	-	2	-	-
PC17. store flammable materials and machine lubricating oil safely and correctly	-	2	-	-
PC18. check that the emission and pollution control devices are working properly in line with environmental policy standards	1	1	-	-
PC19. ensure proper working condition of battery and battery charger	1	1	-	-
PC20. maintain electrolyte level of each cell with distilled water	-	2	-	-
PC21. maintain proper ventilation in battery room	-	1	-	-
PC22. apply good housekeeping practices at all times	1	2	-	-
PC23. identify common hazard signs displayed in various areas	-	2	-	-
PC24. retrieve and/or point out documents that refer to health and safety in the workplace	-	2	-	-
PC25. inform relevant authorities about any abnormal situation/behavior of any equipment/system promptly	-	3	-	-
Fire Safety	4	9	-	-
PC26. use the various appropriate fire extinguishers on different types of fires correctly	1	1	-	-
PC27. distinguish between various types of fire	1	2	-	-
PC28. demonstrate rescue techniques applied during fire hazard	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC29. demonstrate good housekeeping in order to prevent fire hazards	-	2	-	-
PC30. demonstrate the correct use of a fire extinguisher	1	2	-	-
<i>Emergencies, rescue and first-aid procedures</i>	9	21	-	-
PC31. demonstrate how to free a person from electrocution	1	2	-	-
PC32. administer appropriate first aid to victims where required e.g. in case of bleeding, burns, choking, electric shock, poisoning etc.	-	3	-	-
PC33. demonstrate basic techniques of bandaging	1	2	-	-
PC34. respond promptly and appropriately to an accident situation or medical emergency in real or simulated environments	1	2	-	-
PC35. perform and organize loss minimization or rescue activity during an accident in real or simulated environments	1	2	-	-
PC36. administer first aid to victims in case of a heart attack or cardiac arrest due to electric shock, before the arrival of emergency services in real or simulated cases	1	2	-	-
PC37. demonstrate the artificial respiration and the cpr process	1	2	-	-
PC38. participate in emergency procedures emergency procedures: raising alarm, safe/efficient, evacuation, correct means of escape, correct assembly point, roll call, correct return to work	1	2	-	-
PC39. write accident/incident report or dictate a report to another person, and send report to person responsible	1	2	-	-
PC40. demonstrate correct method to move injured people and others during an emergency	1	2	-	-
NOS Total	24	76	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N2001
NOS Name	Use basic health and safety practices for power related work
Sector	Power
Sub-Sector	Generic
Occupation	Generic
NSQF Level	2
Credits	TBD
Version	1.0
Last Reviewed Date	31/03/2022
Next Review Date	14/04/2024
NSQC Clearance Date	31/03/2022

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.



Qualification Pack

Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
PSS/N0101.Assisting in repair and maintenance of power distribution lines and components	24	76	-	-	100	20
PSS/N0102.Preparation and assistance for erection and maintenance of power distribution lines	25	75	-	-	100	20
PSS/N0103.Assisting in operation and maintenance of 11/0.433 KV Distribution Substation	25	75	-	-	100	20
PSS/N0104.Assisting in laying of underground and AB cables	25	75	-	-	100	20
PSS/N1336.Working effectively with others	30	70	-	-	100	5
PSS/N2001.Use basic health and safety practices for power related work	24	76	-	-	100	15
Total	153	447	-	-	600	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
T&D	Transmission and Distribution
REC	Rural Electrification Corporation
AB	Cables Aerial Bunched Cables
HT	High Tension
LT	Low Tension
HV	High Voltage
LV	Low Voltage
BDV	Breakdown Voltage
ULF	Ultra Low Frequency
VLF	Very Low Frequency
OPGW	Optical Groundwire
KV	Kilovolt

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.



Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.